

Domain Knowledge Base: PS26122

Project: Intelligent Data Capture & Schedule-Linking Layer for Infrastructure

Organization: Oil India Limited

1. Project Objective & The Gap

Infrastructure projects map out thousands of micro-tasks in rigid software like Primavera or MS Project. However, execution on the ground is messy. Site supervisors report progress via WhatsApp, paper diaries, and varied spreadsheets.

The Gap: There is a massive terminology and format barrier. A worker writes “spool erected,” but the official schedule expects “Erect Line 24?-XX”. Manual reconciliation takes weeks, making analytics useless.

Our Solution: An LLM-powered “Time Agent” and Fuzzy-Matching backend that ingests messy, conversational field updates, extracts the core intent, and automatically links it to the official L5/L6 master schedule with a confidence score.

2. Core Project Management Vocabulary

Do not use these terms interchangeably in code or presentations.

- **L1 to L6 Schedules:** The hierarchy of a project. L1 is macro (e.g., “Build Oil Rig”). L5/L6 are micro, executable daily activities (e.g., “Weld Pipe A to Valve B”). We are exclusively targeting **L5/L6 nodes**.
 - **Time Agent:** The frontend voice/text interface we are building. It replaces manual forms and lets supervisors just “talk” to the system.
 - **Fuzzy Matching:** The algorithmic process of finding strings that match a pattern approximately (rather than exactly). This bridges the gap between worker slang and official task IDs.
 - **Audit Trail:** The database history. If the AI updates a schedule, it must log exactly *who* said *what* and *when*, so project managers can verify it.
 - **Confidence Score:** A percentage (0-100%) generated by our backend. High scores auto-update the schedule. Low scores are flagged for human review.
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3. System Architecture & Tech Stack Mapping

A. The Data Ingestion Layer (Frontend)

- A responsive web application with a conversational/voice input interface.
- Support for uploading 2–3 varied synthetic formats (e.g., a `.csv` discipline spreadsheet and a free-text daily report).
- *Note: Full production-grade OCR/ASR is not required for the prototype. We will use standard web speech-to-text APIs and text parsers.*

B. The Extraction & Matching Engine (Backend)

- **NLP / LLM Layer:** Extracts the entity (the task), the state (started/finished), and the timestamp from the messy text.
- **String Matching Algorithm:** Implements string manipulation (e.g., Levenshtein distance) to compare the extracted task against an array of thousands of official L5/L6 Primavera task strings.
- **Threshold Logic:**
 - If Match > 85% → Auto-update the master database.
 - If Match < 85% → Flag as “Unmatched/New” and push to a manual review queue. Do *not* silently drop data.

4. Standardized Dashboard UI Matrix

The frontend must render a clean HTML document structure containing the following elements:

Component	Function	Technical Implementation
Input Console	Where supervisors submit updates.	Web forms, text areas, and a microphone API toggle.
Live Audit Table	Shows auto-linked tasks.	HTML table displaying: <code>Extracted Text</code> <code>Matched L5 Node</code> <code>Confidence %</code> <code>Timestamp</code> .
Review Queue	Highlights low-confidence matches.	Ordered/Unordered lists with action buttons (Approve/Reject/Edit) for human managers.

Analytics View	Shows real vs. planned durations.	Visual charts tracking recurring bottlenecks across disciplines (Civil, Piping, Electrical).
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5. Developer Action Items

- **Frontend Team:** Create the HTML/CSS boilerplates. We need a clean dashboard for the “Time Agent” input, a table structure for the Audit Trail, and a Review Queue list.
- **Backend / AI Team:** Research Natural Language Processing (NLP) prompts to extract tasks from sentences. Start building C++ or Python logic for string traversal and fuzzy matching to compare two arrays of text efficiently.
- **Database Setup:** Create a synthetic dataset simulating a Primavera export (JSON/CSV) with 50-100 rigid L5/L6 tasks to act as our “Master Schedule” to test the matching against.